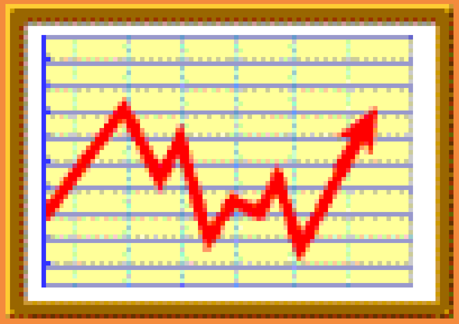
California State University, East Bay
College of Business and Economics

BAN 673 Time Series Analytics

Performance Evaluation and Forecasting Methods

Lecture Materials



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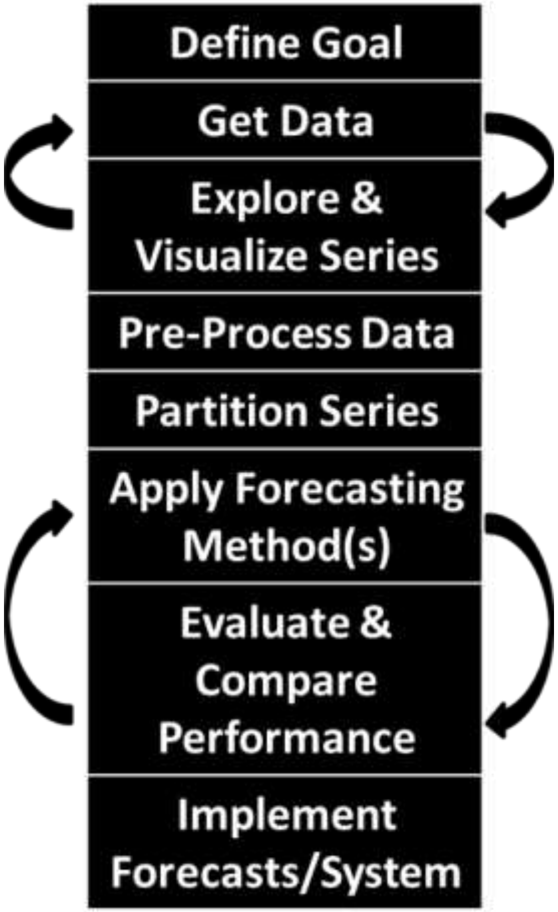
Lecture Objectives

- Identify principles of time series partitioning
- Partition time series data into training and validation data sets
- Recognize model-based and data-driven forecasting methods
- Explain combination of forecasting methods and ensembles
- Identify and explain accuracy measures
- Recognize forecast uncertainty and identify prediction intervals
- Use R to develop time series partitioning, create naïve forecast, measure forecast accuracy, and prediction intervals

Step 5: Partition Time Series

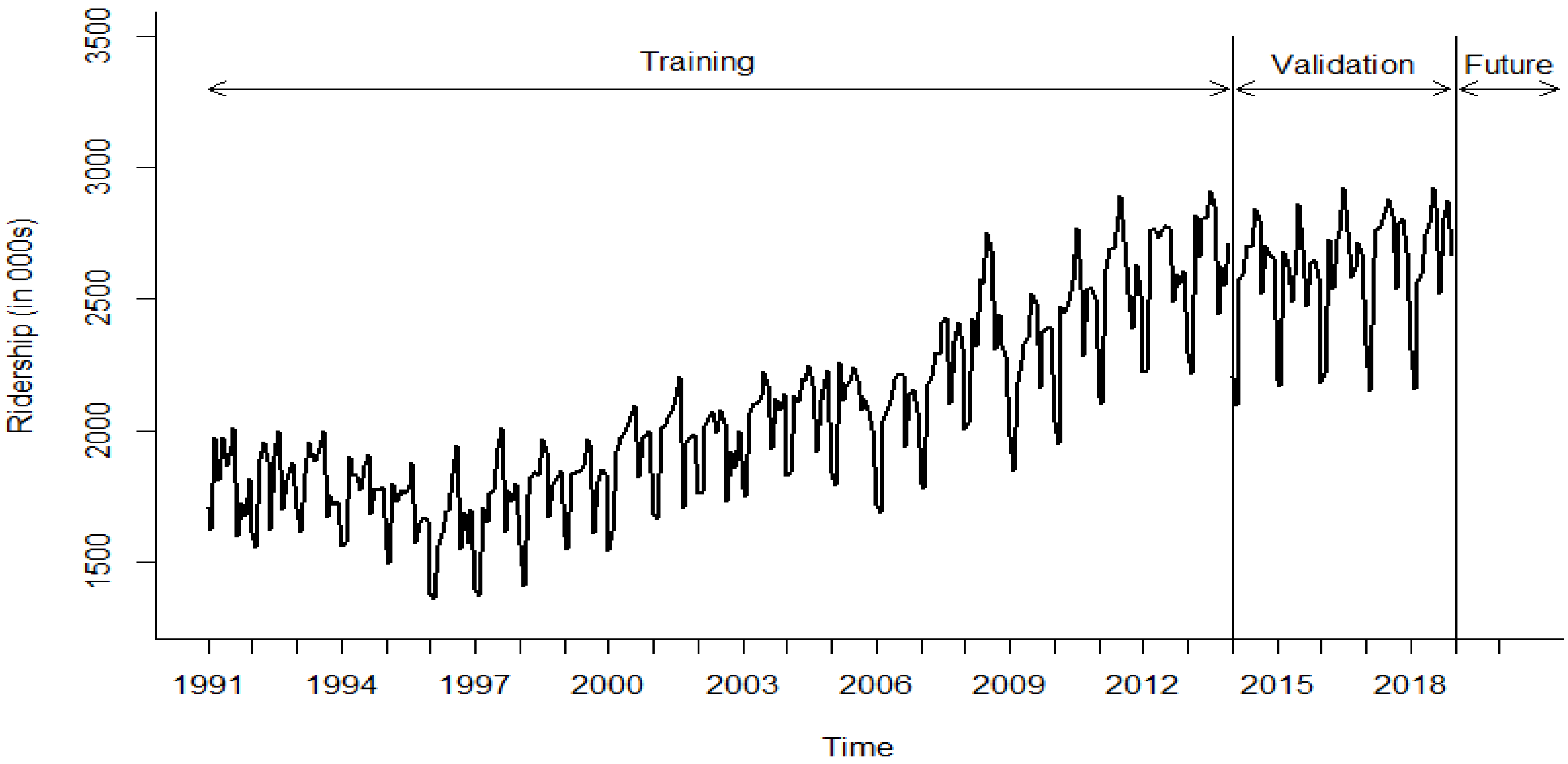
- ***Partition Series*** **is an important preliminary step before applying any forecasting method**
  - **Comes from the need to be able to test how well any selected model performs with the new data not included in the model development**
  - **Typically, divide time series data into two partitions: *training* and *validation* data sets**
  - **Develop a forecasting model based on training data, and test/validate model performance using the validation data**
- **The main reason for partitioning is *overfitting***
  - **It means that a forecasting model is not only fitting the systematic components of the data (e.g., trend and seasonality), but also the noise**
  - ***Overfitted* model is likely to perform poorly on the new data to be forecasted**



Considerations in Data Partitioning

- ***Partitioning of cross-sectional data***
  - **Typically, employs two or three data partitions: *training set*, *validation set*, and sometimes additional *test set***
  - ***Random selection* (in most cases) of data records into training, validation, and test sets**
  - **Training set, the largest in records partition, is used to develop various models, which are assessed by validation and, sometimes, test partitions**
- ***Time series (temporal) partitioning***
  - **The times series is also partitioned into two sets: *training period* and *validation period***
  - ***Nonrandom selection* of partition periods**
    - » **The *earlier period* ($t = 1, 2, \ldots, n$) is designated as the *training period*, the larger partition, typically 70-85% of the whole time series data**
    - » **The *later period* ($t = n+1, n+2, \ldots N$) is designated as the *validation period*, the smaller partition, approximately 15-30% of the whole time series data**
    - » **The proportion of data in training and validation periods can vary depending on time period unit(week, month, quarter), size of the data set, time series patterns**

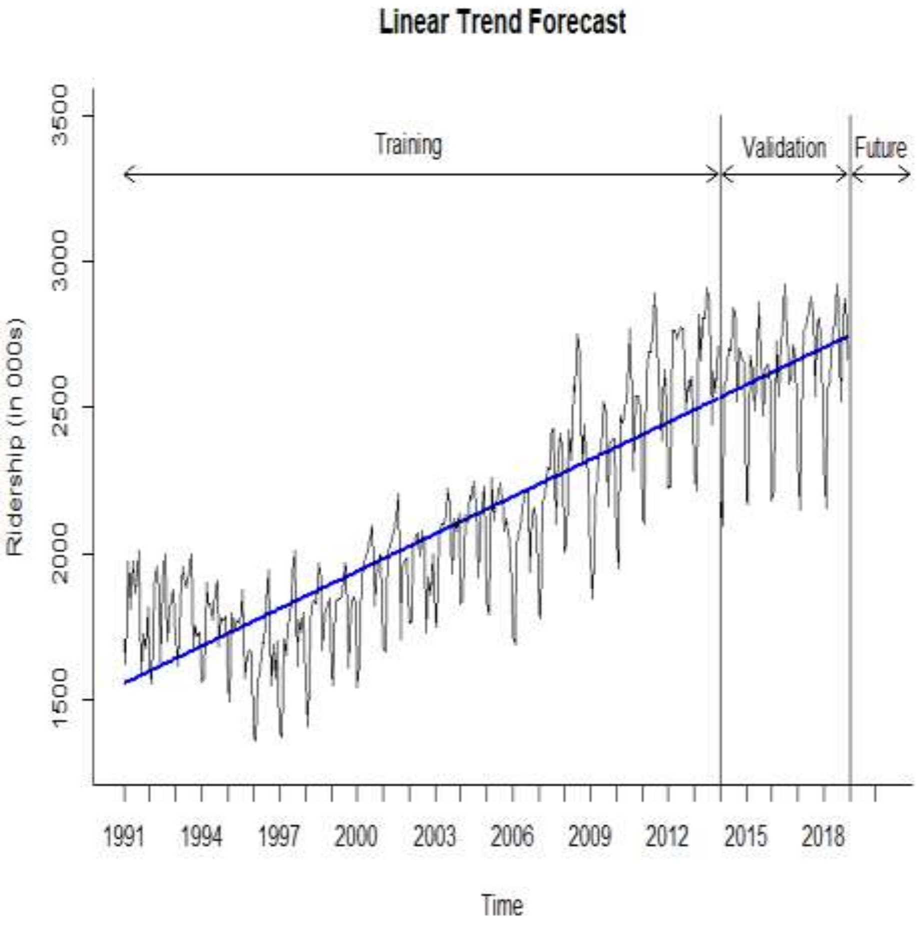
Data Partition Graph For Amtrak

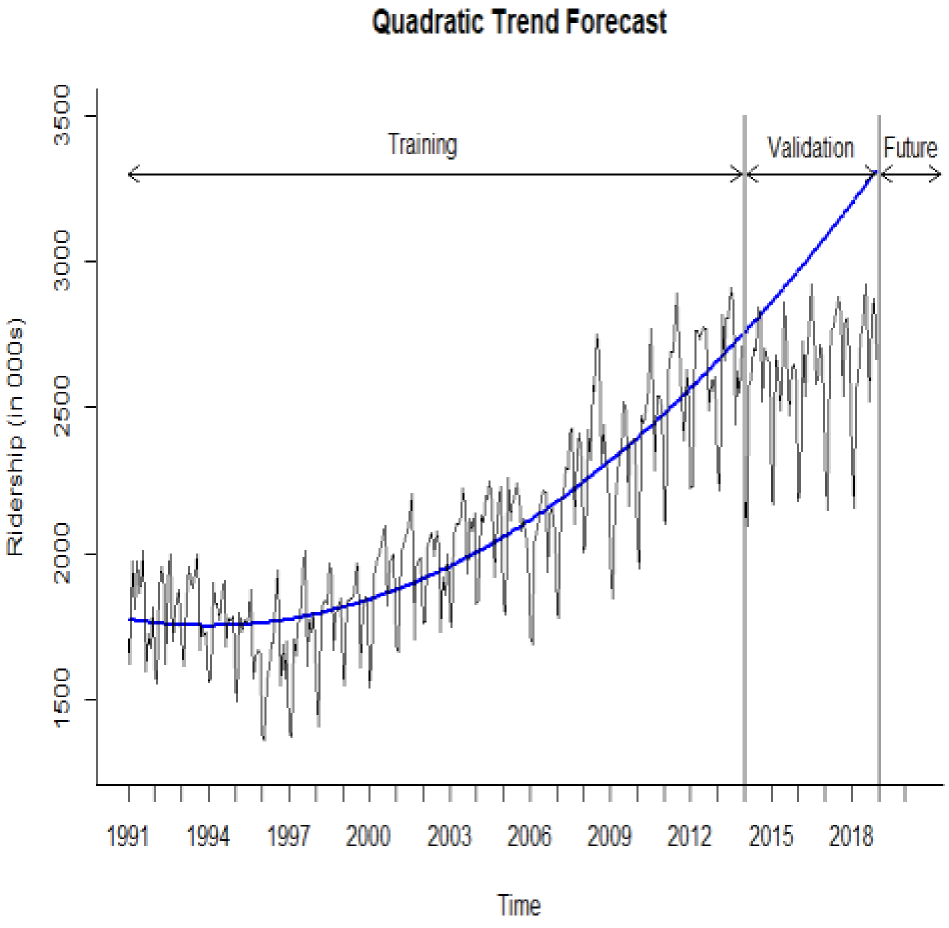


Length of Validation Period and Recombine Partitions for Forecasting

- ***Length of validation period*** **depends on**
  - ***Data frequency*** **(quarterly, monthly, weekly, etc.) and *forecast horizon***
    - » **If the data needs to be forecasted for next year, choose *at least* one year of validation period, e.g., 12 months or 4 quarters**
  - ***Seasonality***
    - » **For monthly or quarterly seasonality, consider validation period of at least 2 years (24 months or 8 quarter, respectively)**
  - ***Length of series***
    - » **The smaller the time series length, the less data will go into validation period**
- ***Recombine partitions for forecasting***
  - **Before attempting to forecast future values of the series, *the training and validation periods must be recombined into one time series*, and *the chosen model needs to be rerun on the complete data***
    - » **The validation period, the most recent one, contains the most valuable information for the future period forecasting, and thus needs to be included for model development**
    - » **With more data for model development, the forecasting model may be estimated more accurately**

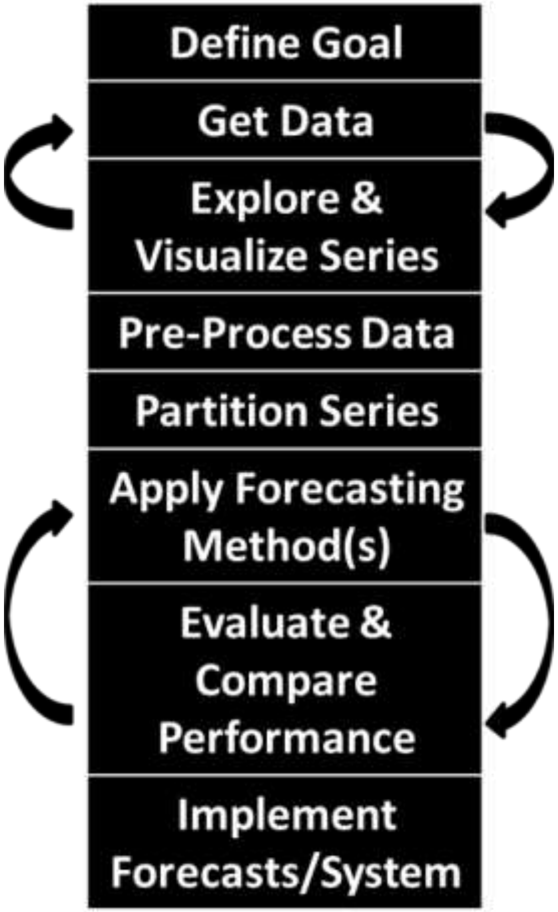
Training and Validation Data with Model Graph





Step 6: Apply Forecasting Method(s)

- *A great variety of forecasting methods*
  - Different methods perform differently depending on the nature of data and forecasting requirements
  - Simple approach like naïve forecast should be considered as a baseline for performance comparison of more sophisticated methods
- Forecasting methods can be roughly divided into *model-based* and *data-driven* methods
- *Model-based methods* apply statistical, mathematical, or other scientific model to forecast time series, especially advantageous for time series with *small number of records*
  - Simple and multiple linear and non-linear regression
  - Autoregressive models
  - ARIMA
  - Logistic regression
  - Econometric models



Forecasting Methods and Level of Automation

- ***Data-driven methods*** **are data dependent and "learn" patterns from data**
- **Advantageous when data structure changes over time and require less user input (then model-based models) in model development**
  - **Naïve forecasts**
  - **Moving average**
  - **Exponential smoothing**
  - **Neural nets**
- ***Manual forecasting***
  - **One-time forecasting**
  - **In-house expertise**
  - **Small number of series to forecast**
  - **Typically, model-based methods**
- ***Automated forecasting***
  - **Ongoing forecasting**
  - **No in-house expertise**
  - **Many series to forecast**
  - **Typically, data-driven methods**
  - **Computationally fast**

Combine Forecasting Methods and Ensembles

- ***Combine multiple forecasting methods*** **to improve predictive performance**
  - ***Two-level (multilevel) methods*, where the first method uses the original time series to predict future, and the second method uses forecast errors from the first method to generate forecast for errors, and then combine two forecasts together**
- ***Ensembles***
  - **Multiple methods applied to the same time series, each generates a separate forecast**
  - **The resulting forecasts are averaged in some way to produce the final forecast**
  - **Can be weighted average, where each forecast has a specific weight proportional to method performance**
  - **Example: ensembles played a major role in Netflix competition to develop the most accurate forecast of movie preference by Netflix customers**

- ***Advantages*** **of combining forecasting methods and ensembles**
  - **More robust forecasts**
  - **Higher precision**
- ***Disadvantages***
  - **Increased cost and time-consuming**
  - **More external consulting expertise**
  - **Requires predetermined rules for combining forecasts**

Naïve Forecast

- ***Naïve forecast*** **is a simple forecasting method that uses the most recent value of the time series**
- **If the actual value of time series in period t is y_t, the naïve k-step ahead forecast, F_{t+k} is**

$$F_{t+k} = y_t$$

- ***Seasonal naïve forecast*** **for a seasonal time series is the value of the most recent identical season.**
  - **For a series with M seasons, seasonal naïve forecast for k-step ahead**

$$F_{t+k} = y_{t-M+k}$$

  - **For example, with monthly time series, the naive forecast for June 2019 will be equal to the actual value in June of 2018**
- **Naïve forecast usage**
  - **Being simple and easy to understand, can sometime be *actually used to achieve useful accuracy levels***
  - **Can be used as a *baseline to compare* it with some other more advanced forecasting methods**

Naïve Forecast Calculations

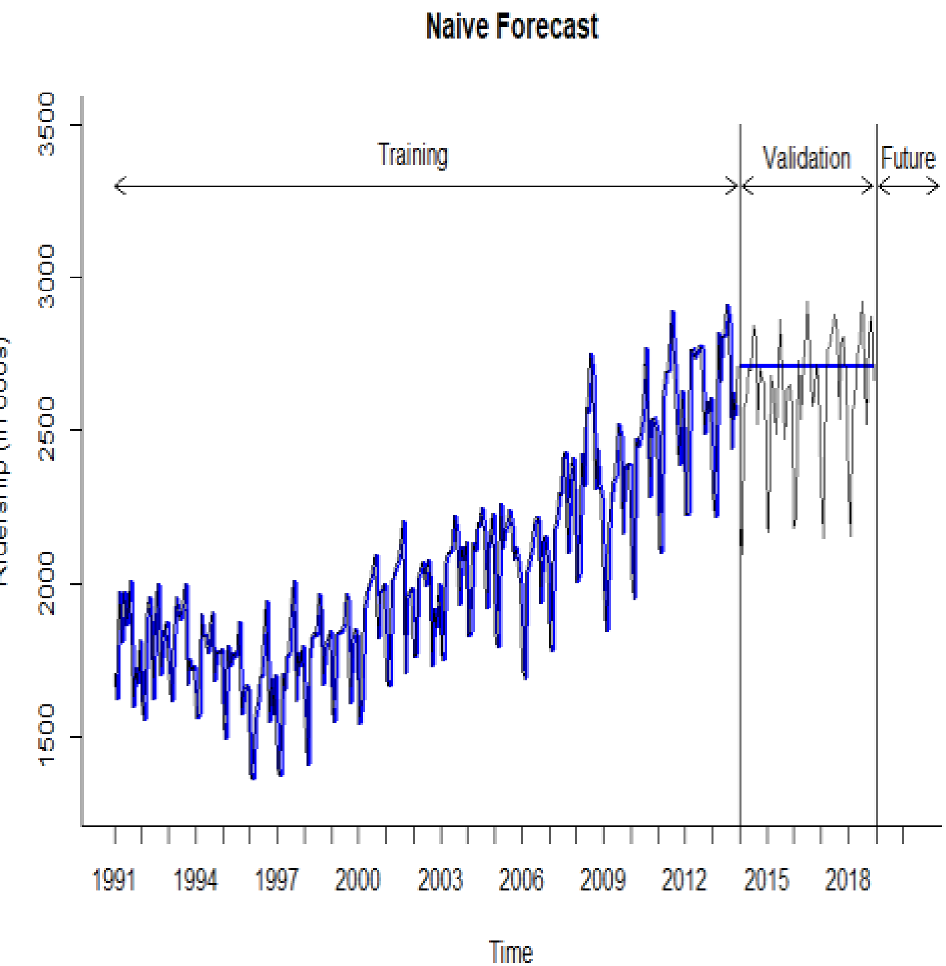
Month	*Ridership*	*Naïve Forecast*	*Seasonal Naïve Forecast*
1/1/1991	1708.917	#N/A	#N/A
2/1/1991	1620.586	1708.917	#N/A
3/1/1991	1972.715	1620.586	#N/A
4/1/1991	1811.665	1972.715	#N/A
5/1/1991	1974.964	1811.665	#N/A
6/1/1991	1862.356	1974.964	#N/A
7/1/1991	1939.860	1862.356	#N/A
8/1/1991	2013.264	1939.860	#N/A
9/1/1991	1595.657	2013.264	#N/A
10/1/1991	1724.924	1595.657	#N/A
11/1/1991	1675.667	1724.924	#N/A
12/1/1991	1813.863	1675.667	#N/A
1/1/1992	1614.827	1813.863	1708.917
2/1/1992	1557.088	1614.827	1620.586
3/1/1992	1891.223	1557.088	1972.715
4/1/1992	1955.981	1891.223	1811.665
5/1/1992	1884.714	1955.981	1974.964
6/1/1992	1623.042	1884.714	1862.356
7/1/1992	1903.309	1623.042	1939.860
8/1/1992	1996.712	1903.309	2013.264
9/1/1992	1703.897	1996.712	1595.657
10/1/1992	1810.000	1703.897	1724.924
11/1/1992	1861.601	1810.000	1675.667
12/1/1992	1875.122	1861.601	1813.863

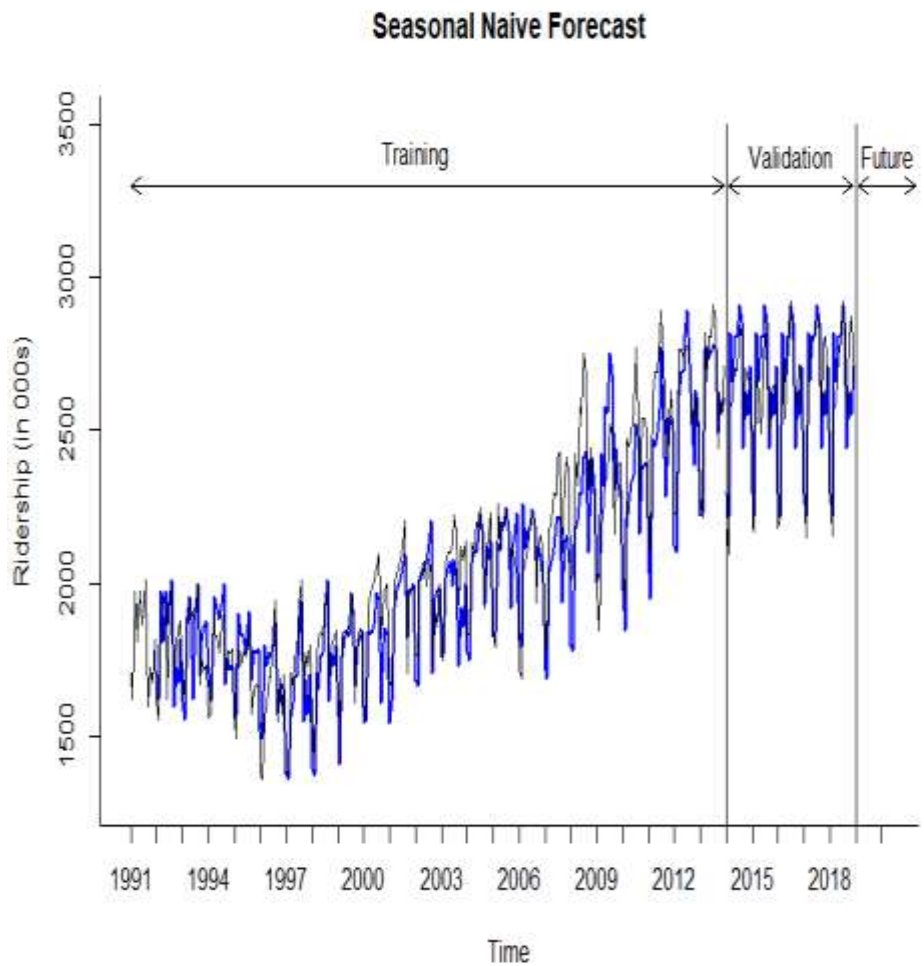
Naïve/seasonal naïve forecast for training period

Month	*Ridership*	*Naïve Forecast*	*Seasonal Naïve Forecast*
1/1/2013	2298.047	2604.566	2224.002
2/1/2013	2217.348	2298.047	2231.641
3/1/2013	2816.139	2217.348	2763.361
4/1/2013	2659.243	2816.139	2763.894
5/1/2013	2803.527	2659.243	2735.455
6/1/2013	2804.717	2803.527	2757.916
7/1/2013	2908.19	2804.717	2777.117
8/1/2013	2851.979	2908.190	2769.229
9/1/2013	2440.153	2851.979	2487.441
10/1/2013	2625.82	2440.153	2596.272
11/1/2013	2550.704	2625.820	2559.615
12/1/2013	2711.851	2550.704	2604.566
1/1/2014	2206.788	2711.851	2298.047
2/1/2014	2092.819	2711.851	2217.348
3/1/2014	2575.951	2711.851	2816.139
4/1/2014	2592.994	2711.851	2659.243
5/1/2014	2700.179	2711.851	2803.527
6/1/2014	2695.954	2711.851	2804.717
7/1/2014	2844.949	2711.851	2908.190
8/1/2014	2802.873	2711.851	2851.979
9/1/2014	2519.583	2711.851	2440.153
10/1/2014	2702.607	2711.851	2625.820
11/1/2014	2667.575	2711.851	2550.704
12/1/2014	2656.185	2711.851	2711.851
1/1/2015	2193.845	2711.851	2298.047
2/1/2015	2166.654	2711.851	2217.348
3/1/2015	2676.17	2711.851	2816.139

Naïve/seasonal naive forecast for validation period

Naïve Forecast for Amtrak Data



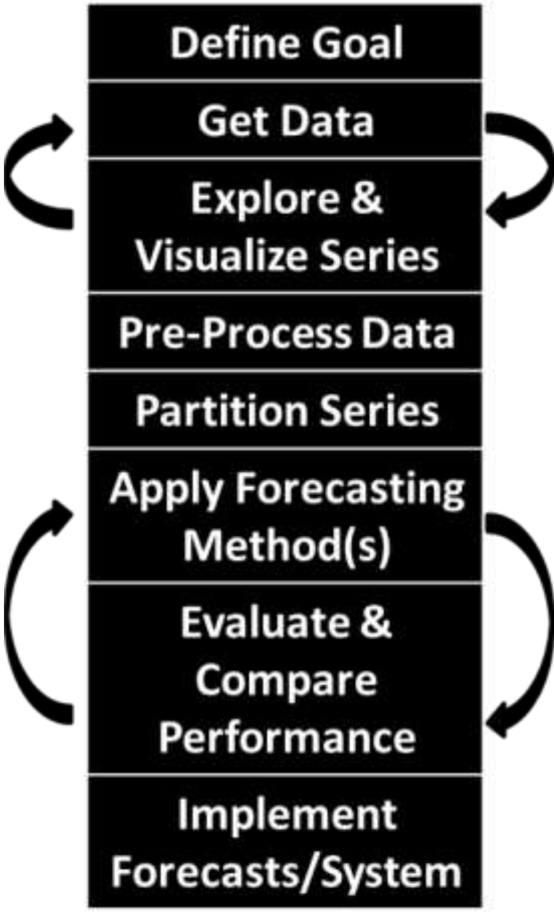


Step 7: Evaluate & Compare Performance

- ***Performance measures*** are used in forecasting to evaluate prediction accuracy and compare accuracy of various forecasts to identify the best forecast
- ***Forecast accuracy*** refers to how close forecasts come to actual data
- ***Accuracy*** is usually measured in forecasting by the value of its adverse characteristic--forecast error
- ***Forecast error or residual*** (e_t) is the difference between actual (y_t) and forecasted (F_t) values in the same period

$$e_t = y_t - F_t$$

- The smaller the forecast error, the more accurate the forecast
- Performance measures are typically based on ***validation period data***



Common Accuracy Performance Measures

- **Consider a validation period with *v* periods (records), *t = 1, 2, …, v*, and error e_t (difference between actual and forecasted values) in each respective period**
- ***Mean/average error (ME or AE)*** **– provides an indication of whether the forecast is over- or under-predicting**

$$ME = \frac{1}{v}\sum_{t=1}^{v} e_t$$

- ***Mean absolute error/deviation (MAE or MAD) give the magnitude of the average absolute error:***

$$MAE = \frac{1}{v}\sum_{t=1}^{v} |e_t|$$

- ***Mean absolute percentage error (MAPE)*** **gives an absolute percentage score of how forecast deviates (on the average) from actual values; useful for comparing performance across series of data that have different scales:**

$$MAPE = \frac{100}{v}\sum_{t=1}^{v} \left|\frac{e_t}{y_t}\right|$$

- ***Root mean square error (RMSE)*** **has the same units as the time series and sensitive to large errors**

$$RMSE = \sqrt{\frac{1}{v}\sum_{t=1}^{v} {e_t}^2}$$

Calculation of Accuracy Measures in Excel

Validation Data and Linear Trend Forecast

Month	Ridership (Actual)	Forecast	Error (e_t)	Absolute Error ($\|e_t\|$)	Absolute Proportion of Error ($\|e_t/y_t\|$)	Squared error (e_t^2)
1/1/2014	2206.788	2534.333	-327.545	327.545	0.148	107285.766
2/1/2014	2092.819	2537.877	-445.058	445.058	0.213	198076.368
3/1/2014	2575.951	2541.420	34.531	34.531	0.013	1192.365
4/1/2014	2592.994	2544.964	48.030	48.030	0.019	2306.879
5/1/2014	2700.179	2548.508	151.671	151.671	0.056	23004.192
6/1/2014	2695.954	2552.051	143.903	143.903	0.053	20707.981
7/1/2014	2844.949	2555.595	289.354	289.354	0.102	83725.752
8/1/2014	2802.873	2559.139	243.734	243.734	0.087	59406.444
9/1/2014	2519.583	2562.682	-43.099	43.099	0.017	1857.548
10/1/2014	2702.607	2566.226	136.381	136.381	0.050	18599.796
11/1/2014	2667.575	2569.770	97.805	97.805	0.037	9565.899
12/1/2014	2656.185	2573.313	82.872	82.872	0.031	6867.729
1/1/2015	2193.845	2576.857	-383.012	383.012	0.175	146698.108
2/1/2015	2166.654	2580.401	-413.747	413.747	0.191	171186.201
3/1/2015	2676.17	2583.944	92.226	92.226	0.034	8505.599
4/1/2015	2621.089	2587.488	33.601	33.601	0.013	1129.037
5/1/2015	2487.546	2591.031	-103.485	103.485	0.042	10709.249
6/1/2015	2683.762	2594.575	89.187	89.187	0.033	7954.294
7/1/2015	2861.747	2598.119	263.628	263.628	0.092	69499.826
8/1/2015	2692.783	2601.662	91.121	91.121	0.034	8302.953
9/1/2015	2472.007	2605.206	-133.199	133.199	0.054	17742.003
10/1/2015	2636.686	2608.750	27.936	27.936	0.011	780.433
11/1/2015	2648.743	2612.293	36.450	36.450	0.014	1328.572
12/1/2015	2582.766	2615.837	-33.071	33.071	0.013	1093.695

ME	MAE	MAPE	RMSE
-22.335	159.359	6.472	208.788

Additional Accuracy Performance Measures

- ***Mean percentage error (MPE)*** **is similar to MAPE but without absolute value of the ratio; indicates the percentage of over- or under-predicting**
- ***Mean absolute scaled error (MASE)*** **is a scaled version of MAE, which identifies the ratio of validation MAE vs. average naïve forecast error**
  - **If *MASE < 1* – forecasting model has a lower average error than naïve forecast**
  - **If *MASE ≥ 1* – forecasting model has the same or higher average error than naïve forecast**

$$MASE = \frac{Validation\ MAE}{Training\ MAE\ of\ Naive\ Forecasts}$$

- ***ACF1*** **is autocorrelation (autocorrelation coefficient) of errors at lag 1**
- ***Theil's U statistic*** **is a relative accuracy measure that compares the forecasted results with the results of forecasting with minimal historical data**
  - **If *U < 1* – forecasting model is better than guessing**
  - **If *U = 1* – forecasting model is as good as guessing**
  - **If *U > 1* – forecasting model is worse than guessing**

Performance Measures Using Accuracy Function in R

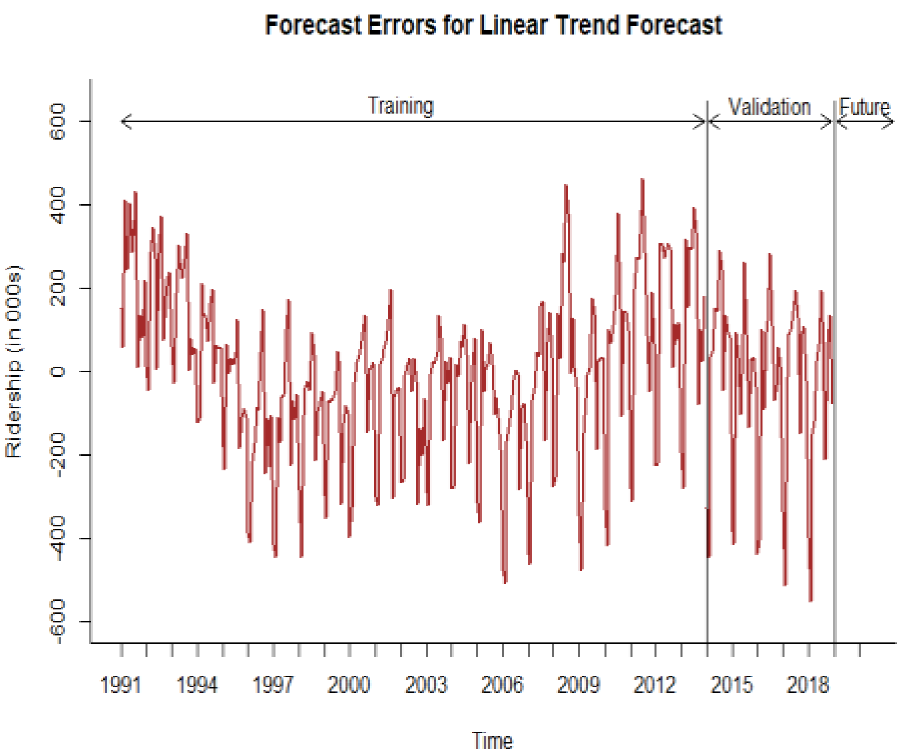
Regression with Linear Trend – Validation Data

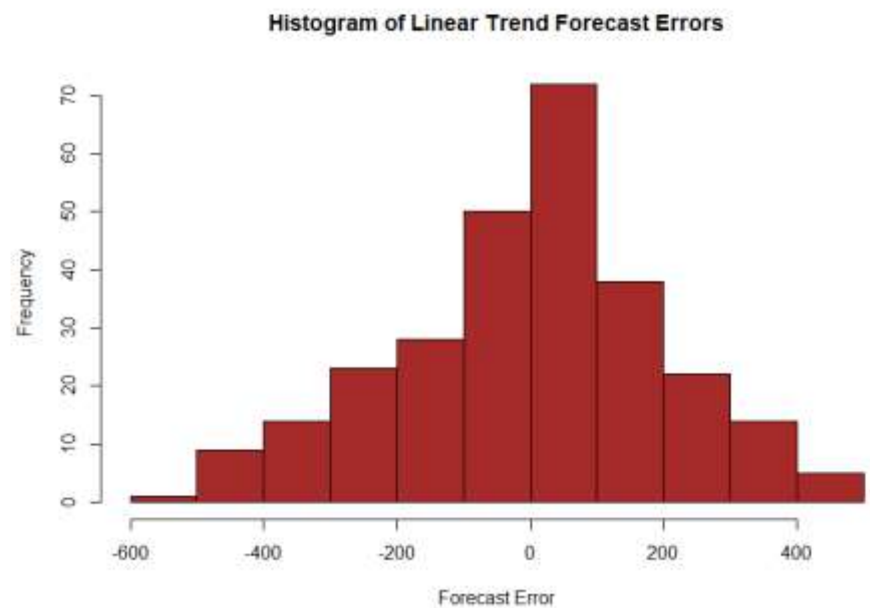
	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-22.335	208.788	159.359	-1.553	6.472	0.431	0.919

Regression with Quadratic Trend – Validation Data

	ME	RMSE	MAE	MPE	MAPE	ACF1	Theil's U
Test Set	-408.405	470.314	409.616	-16.33	16.372	0.553	2.045

Error Graph and Histogram in R





Measure Forecast Uncertainty

- ***Prediction (confidence) interval*** **is a forecast range for a forecasted period, which has an uncertainty level attached to it**
  - **Based on a probability (typically 95%; in R – default to 80% and 95%) that the forecasted values will be in the range [a, b]**
  - **More informative than a single forecast number**
  - **Tell about uncertainty associated with the forecast similar to MAPE but without absolute value of the ratio; indicates the percentage of over- or under-predicting**
- ***With normally distributed errors*****, prediction interval for forecast F_t:**

$$F_t \pm Z\sigma,$$

where σ **= estimated standard deviation of forecast errors**

Z **= Z-score associated with probability(confidence level)**

(*Z = 2* corresponds with *95%* confidence level probability)

- **For non-normal errors, *transform errors to normal***

Forecast with Prediction (Confidence) Interval